Chaomeng Lu / 卢超猛

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Summary

PhD researcher in Computer Science at KU Leuven (DistriNet), specializing in AI-driven software security and vulnerability detection. First author of papers in the Journal of Systems and Software, IEEE/ACM MSR, and IEEE Internet of Things Journal; builder of open-source vulnerability datasets and evaluation frameworks for deep learning models and LLMs. Seeking engineering or research roles in AI, software engineering, and cybersecurity, with a focus on turning research methods into reliable, production-oriented solutions.

Education

Dec 2023 – Sep 2027 **PhD in Computer Science**, KU Leuven *(expected)*
Thesis topic: Deep Learning and LLM-based Automated Vulnerability Detection in Source Code

Sep 2020 – Jun 2023 **Master in Cyberspace Security**, North China University of Science and Technology
Thesis: Research on Network Intrusion Data Analysis Based on Meta-Learning

Sep 2016 – Jun 2020 **Bachelor in Network Engineering**, North China Institute of Science and Technology

Experience

PhD Researcher, DistriNet, KU Leuven Dec 2023 – Present
Leuven, Belgium

- Research AI-driven vulnerability detection in C/C++ source code using pre-trained code models, contrastive learning, and LLMs.
- Built vulnerability data pipelines (CVEs and fix commits from 800+ GitHub repositories, SZZ-traced contributing commits) and reproducible benchmarks for deep learning models and LLMs.
- Published in JSS and MSR; released code and datasets as open-source artifacts; serve on the MSR program committee.

Data Intern, ZF Group May 2023 – Aug 2023
Shanghai, China

- Processed and cleaned internal business datasets to support reporting and operational analysis.
- Built data visualizations and dashboards to communicate trends, anomalies, and key metrics to stakeholders.
- Transformed raw data into structured, analysis-ready formats using Python, Excel, and VBA, improving reporting workflows.

Publications and Projects

- **PatchCon: Patch-aware Weighted Supervised Contrastive Learning for Vulnerability Detection**
  arXiv preprint (under review), 2026 Paper | Project
  Proposed PatchCon, a CodeBERT-based model that jointly optimizes vulnerability classification and supervised contrastive learning, treating vulnerable–patched function pairs as weighted hard negatives to capture subtle fix-related differences. On PairVul-Linux, improved pairwise accuracy over the strongest reproduced baseline from 0.21 to 0.33 (+58%), matched a GPT-4o-based LLM method at substantially lower cost, and generalized best to unseen CVEs and newly disclosed Linux kernel vulnerabilities.

- **From Lab to Reality: A Practical Evaluation of Deep Learning Models and LLMs for Vulnerability Detection**
  Journal of Systems and Software (JSS), accepted, 2026 Paper | Project
  Conducted a deployment-oriented evaluation of deep learning models and open-weight LLMs for source-code vulnerability detection across multiple benchmark datasets, and assessed their real-world effectiveness on recent Linux kernel and Chromium vulnerabilities.

- **ICVul: A Well-labeled C/C++ Vulnerability Dataset with Comprehensive Metadata and VCCs**
  IEEE/ACM 22nd International Conference on Mining Software Repositories (MSR), 2025 Paper | Project
  Built ICVul, a C/C++ vulnerability dataset derived from real-world CVEs and GitHub fix commits across 800+ repositories, enriched with function-, file-, and commit-level metadata and Vulnerability-Contributing Commits traced using SZZ.

– **A Few-shot Based Model-Agnostic Meta-Learning for Intrusion Detection in Security of Internet of Things**
IEEE Internet of Things Journal, vol. 10, no. 24, 2023 Paper | Project
Developed a few-shot IoT intrusion detection approach based on model-agnostic meta-learning to improve adaptation to new attack types with limited labeled samples.

– **Application of Meta-learning in Cyberspace Security: A Survey**
Digital Communications and Networks, vol. 9, no. 1, 2023 Paper (co-author)
Reviewed how meta-learning addresses few-shot, zero-shot, and unknown-attack challenges in cyberspace security, summarizing its main model types, application areas, and future prospects.

– **A Comparative Analysis of Vulnerability Management in Machine Learning Projects**
IEEE EuroS&P 2026 Workshops (DeMeSSAI), Lisbon, 2026 (co-author)

– **A Multi-dimensional Visual Analytics Tool for the Security Posture of Open-source Software**
IEEE/ACM 3rd International Workshop on Software Vulnerability Management (SVM), 2025 (co-author)

– **Self-hosted MLflow Tracking Platform on Azure**
Deployed a self-hosted MLflow Tracking Server on an Azure Ubuntu VM with Nginx reverse proxy, Basic Auth, systemd service management, and artifact tracking for centralized experiment management.

Skills

Programming	Python, C/C++, Java, SQL, Bash
Machine Learning	PyTorch, TensorFlow, Transformers, vLLM, scikit-learn, LLMs, Pre-trained Code Models (CodeBERT), Contrastive Learning, Meta-Learning, NLP, HPC
Security	Software Vulnerability Detection, Static Code Analysis, Software Security, Network Intrusion Detection
Data Engineering	Pandas, NumPy, Data Cleaning and Labeling, Mining Software Repositories, Statistical Analysis, Data Visualization
MLOps / Cloud	MLflow, Docker, Git, Linux, Azure, MySQL, PostgreSQL, Nginx

Languages

English	Full professional proficiency
Chinese	Native proficiency
Dutch	Elementary proficiency

Professional Service

Program Committee Member, MSR 2027 Data and Tool Showcase Track 2026
Program Committee Member, MSR 2026 Data and Tool Showcase Track 2025

Awards & Academic Activities

– **Second Prize**, Network and Information Security Technology Competition (CTF), Hebei Department of Education, 2021

– **Third Prize**, "Blue Bridge Cup" National Software Professionals Design Contest, 2018

– **Attendee**, 11th IEEE European Symposium on Security and Privacy (EuroS&P), Lisbon, Jul 2026

– **Attendee**, 47th IEEE/ACM International Conference on Software Engineering (ICSE), Ottawa, Apr–May 2025

– **Attendee**, 22nd International Conference on Mining Software Repositories (MSR), Ottawa, Apr–May 2025

– **Participant**, Summer School on Security and Privacy in the Age of AI, KU Leuven, Sep 2024

Interests

Running, hiking, cycling, guitar, music